starts with 35 and no digit appear more than once?

4. Find the number of arrangements that can be made out of the letters of the word "ASSASSINATION"

5. (a) In how many ways can 8 identical beads be strung on a necklace?
   (b) In how many ways can 8 boys form a ring?

6. Find the rank of the word 'CHAT' in dictionary.

2.3 Combinations

A combination is a selection of items from a collection such that the order of selection does not matter. That is the act of combining the elements irrespective of their order. Let us suppose that in an interview two office assistants are to be selected out of 4 persons namely A, B, C and D. The selection may be of AB, AC, AD, BC, BD, CD (we cannot write BA, since AB and BA are of the same selection). Number of ways of selecting 2 persons out of 4 persons is 6. It is represented as $4C_2 = \frac{4\times3}{1\times2} = 6$.

The process of different selection without repetition is called Combination.

Definition 2.4

Combination is the selection of n things taken r at a time without repetition. Out of n things, r things can be selected in nc_r ways.

$$nC_r = \frac{n!}{r!(n-r)!},\ n \geq 1,\ 0 \leq r \leq n$$

Here $n \neq 0$ but r may be 0.

For example, out of 5 balls 3 balls can be selected in $5C_3$ ways.

$$5C_3 = \frac{5!}{3!(5-3)!} = \frac{5!}{3!\,2!} = \left(\frac{5\times4\times3}{1\times2\times3}\right)$$
$$= 10 \text{ ways.}$$

Example 2.25

Find $8C_2$

Solution

$$8C_2 = \frac{8\times7}{2\times1} = 28$$

DO YOU KNOW?

Permutations are for lists (order matters) and combinations are for groups (order doesn't matter)

Properties

(i) $nC_0 = nc_n = 1$

(ii) $nC_1 = n$

(iii) $nC_2 = \frac{n(n-1)}{2!}$

(iv) $nC_x = nC_y$, then either $x = y$ or $x + y = n$

(v) $nC_r = nC_{n-r}$

(vi) $nC_r + nC_{r-1} = (n+1)C_r$

(vii) $nC_r = \frac{np_r}{r!}$

Example 2.26

If $nC_4 = nC_6$, find $12C_n$.

Solution

If $nC_x = nC_y$, then $x + y = n$.

Here $nC_4 = nC_6$

$\therefore n = 4 + 6 = 10$

$12C_n = 12C_{10}$

$= 12C_2$

$= \frac{12\times11}{1\times2} = 66$

Example 2.27

If $nP_r = 720$; $nC_r = 120$, find r

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